



EG1113 - C Programming for Engineering Lab

Report #12

Race Day

Date of the experiment: November 20, 2024

Due date of report: November 22, 2024

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1. Core concepts learned

Summarize the core concepts learned and practiced in this laboratory.

- Create efficient code to drive a car through a course
- Have fun

2. List of equipment and software used

- Raspberry PI 400
- Notepad
- Lenovo Laptop
- Caja File Network Share

3. Relevant web links and other documents:

- Non were utilized

4. Race

4.1 Summary

- Remove all warnings from code
- Ability to trim vehicles
- Code Modifications
- Race Day Videos (Fastest time 30.28 s) : [Race Day](#)

4.2 Procedures

1. Trim Cars, nothing fancy just hard coded in

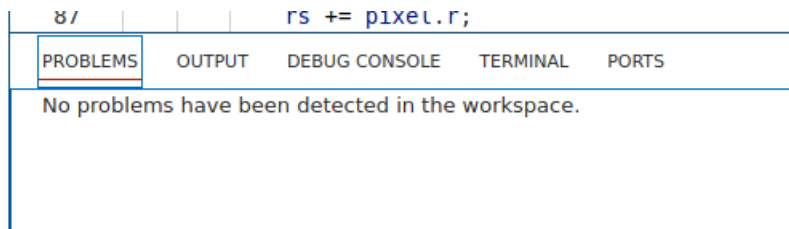
```
1 #include "car_trim.h"
2 // Percent to trim the car so that it will go straight based on each differnt button value zero through fifteen
3 //
4 // Value based on button values described in comments below (lf=left, rt=right, fw=forward, str=straight, bk=back)
5 //
6 //
7 //      off    lf fw    lf bk    -    rt bk    spin lf    bk str    -    rt fw    fw str    spin rt    -    -    -    -    -
8 //      0      1      2      3      4      5      6      7      8      9      10     11     12     13     14     15
9 int trim_percent[4][16]={
10 { 0, 0, 100, 0, 0, 0, 100, 0, 0, 0, 0, 100, 0, 0, 0, 0},
11 { 0, 0, 0, 0, 0, 100, 100, 100, 0, 0, 0, 0, 0, 0, 0, 0},
12 { 0, 0, 0, 0, 0, 0, 0, 0, 0, 100, 100, 100, 0, 0, 0, 0}};
13 // array of current duty cycles for each pin -- most will be zero, but we don't want to start the PWM on a pin if is already running because it could cause an anomaly
```

2. Modify Code

2.1 Only displays camera by disabling virtual buttons, centroid, and initializing window to 640 x 480

```
161
162 BeginDrawing();
163 //ClearBackground(BLUE);
164 //DrawRectangle(50 + 800 / 2 - 250 / 2, 50, 150, 150, WHITE);
165 do draw_frame(frame_x, frame_y);
166 //DrawCircle(800 / 2 - 250 / 2 + 75, 75, 18, buttons & 8 ? CLITERAL(Color){255, 31, 35, 255 * current_speed / 100} : GRAY);
167 //DrawCircle(800 / 2 - 250 / 2 + 250 - 75, 75, 18, buttons & 1 ? CLITERAL(Color){255, 31, 35, 255 * current_speed / 100} : GRAY);
168 //DrawCircle(800 / 2 - 250 / 2 + 75, 250 - 75, 18, buttons & 4 ? CLITERAL(Color){0, 228, 48, 255 * current_speed / 100} : GRAY);
169 //DrawCircle(800 / 2 - 250 / 2 + 250 - 75, 250 - 75, 18, buttons & 2 ? CLITERAL(Color){0, 228, 48, 255 * current_speed / 100} : GRAY);
170
171 //update_and_draw_sprites();
172 EndDrawing();
173
```

3. Ensure no warnings



4. Race the cars

5. Problems experienced and areas of concern

- No noteworthy ones

6. Race Improvements

- Livestreaming the race
- Running the code locally on the cars
- Having the option to bring in your own car (3D Printed)
- Final race, anything goes(hacking), or having two races to eliminate possible advantages

7. Course goals

Rank the level (0=none to 5=a large amount) that this lab covered the each of the course goals by updating the appropriate box in the table below.

Course Goals	Laboratory												
	01	02	03	04	05	06	07	08	09	10	11	12	13
To compile, link, debug and run C programs	4	5	5	5	5	5	5	5	0	0	5	5	
To use the basic C program structure; variables, constants, and operators	4	4	4	4	4	5	5	5	0	0	5	5	
To use the repetition constructs such as looping with for, while, and do-while statements	2	3	4	4	4	4	4	4	0	0	5	5	
To use the selection structures such as if, if/else, switch, conditional expression statement	0	0	3	3	3	3	4	4	0	0	5	5	
To create program modules using functions (passing data to and returning values from functions)	0	0	0	2	4	4	4	4	0	0	5	5	
To use arrays and pointers	0	0	0	0	4	3	4	4	0	0	5	5	
To work with strings and string manipulating functions	2	4	4	1	2	1	2	4	0	0	5	5	
To perform file, I/O operations	3	4	4	5	5	5	5	5	0	0	5	5	
To develop structured, modular, and top-down design of software	0	0	0	1	1	4	4	4	0	0	5	5	